THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NORTH WESTERN ZONE FORM FOUR FORM PRE-NATIONAL EXAMINATION

033/1

TIME: 3HRS BIOLOGY 19/9/2024

INSTRUCTIONS

1. This paper consists of sections A, B and C
2. Answer all questions in sections A and B, and two questions from section C.
3. All drawings must be in pencil and writings must be in blue or black pen.
4. Cellular phones, watches and all unnecessary materials are not allowed in the examination room.

**SECTION A (16 MARKS)**
**Answer all questions**

**1**. In question (i)-(x) select the best answer among the given alternatives and write its letter in the space provided

( i ) Students at Mtakuja secondary were educated about risk behaviors that must be abstained. the behaviors to be abstained were:

A. Alcoholism and patience

B. Unsafe sexual intercourse and exercising

C. Raping and alcoholism

D. Sharing books during studying

E. Patience and exercising

(ii) In a garden of pea, a gardener observed the seedling with its cotyledons on the ground. Those seedlings undergo:

A. Positive growth B. Epicotyl germination Hypogeal germination

D. Epigeal germination E. hypocotyl germination

(iii ) In order to avoid dislocation during movement, two bones are held together by strong tissue called

A. Muscles. B. Tendons. C. Cartilage. D. ligaments. E. fibres

(iv) When students were doing private study at night, light went off suddenly. Which of the following change occurred in the eyes of the students?

A. The lens become thicker B. The eyes opened wider C. The iris enlarged D. The pupil enlarged Either eyelids closed

(v) During any mitotic division .....number of cells is expected to be produced

A. Four B. One C Two D. Three E. Five

(vi)Which among the following is not an infection and disease of the respiratory system?

A. Asthma B. Bronchitis C. Pneumonia D. Emphysema E. Leukemia

(vii) What among of the following conditions or factors can be changed or change in an experiment?

A. Problem B. Hypothesis. C Conclusion D. variables E. Observations

(viii) One of the following is the similarity between division coniferophyta and division angiospermophyta:

A. They reproduce sexually by seed B. They are wind pollinated C. They have cones D. They have seeds with cotyledons E. they produce flowers

(ix)Chickens and dogs have the ability to maintain constant body temperature. They are described as:

A. Homeotherms B. Poikilotherms C.Ectotherms D. Hibernators E. Regulators

(x) Johntel started salivating as he saw peeled half ripe mango with topping of red chili powder. What part of the brain is concerned with Johntel's salivation?

A. Hypothalamus B. Medulla oblongata C. Cerebellum D. Cerebrum. E. Cerebral hemisphere

**2**. Match the description of list A with the corresponding responses in list B by writing the letter of the correct response besides the item number in the answer sheet provided

| LIST A | LIST B |
|---|---|
| i)transformation of a seed embryo into a seedling<br>ii)An opening that allows the entry of water and oxygen into a seed during germination process<br>iii)Growth ceases when a certain age is reached<br>iv)Internal factors necessary for seed germination<br>v)A period of reduced metabolic activities in the seed<br>vi) In hibis the growth of lateral buds during plant growth | A. Determinate growth<br>B. Indeterminate growth<br>C. Apical dominance<br>D. Hilum<br>E. Micropyle<br>F. Seed dormancy<br>G. seed germination<br>H. enzyme and hormones<br>I. Optimum temperature ,moisture and oxygen |

## SECTION B (54 MARKS)
**Answer all Questions from this section**

**3**. There is a very big debate to why Biology should not be taken as an optional subject like other optional subjects at ordinary secondary school. You as an expert in Biology state reasons to why Biology should be studied by all O-level students (Give six points)

**4**. Water and salts are needed in constant optimal amount for effective and proper functioning of the body.

(a) Explain three factors that affect the content of water and salt in the body
(b) Mention the two hormones which are responsible for regulation of water and salt in the human body

**5**. (a)Explain how a plant cell prevents bursting when placed in hypotonic solution
(b)What will be the effect in a recipient's blood when he or she has been transfused an Incompatible blood from the donor?
(c)State any three precautions to be taken before blood transfusion

**6**. Around the school farm there were two kinds of flowers classified as group A and group Group A flowers attract insects so much and group B were not. The two kinds undergo different types of pollination.
(a)Identify the types of pollination favorable for group A and B
(b) Differentiate flowers in group A from those in group B

**7**. In order for organisms to grow well there must be a particular kind of cell division
(a) Identify and give the meaning of the kind of cell division responsible for growth
(b) How is the kind of cell division in 7(a) above significant? Give 4 points

**8.** (a) State Mendel's first law of inheritance.
(b)Give any four differences between DNA and RNA
(c) Beam seed placed in a closed bottle containing cold water, swollen and started rotting in steady of germinating. Why?

## SECTION C (30 MARKS)
## Answer two questions from this section

**9.** Imagine you are a guest speaker on the workshop on Birth control methods at Kazaroho Village, educate the workshop participants on birth control methods which:
(a) Prevent the union of gametes (4 points)
(b) Suppress the formation or the release of gametes (two points)
(c) Prevent implantation of the zygote (1 point)

**10**. The body of the human being is made up of frame work of interconnected bones which play crucial roles to the body. Justify this statement (4points).

**11**. Suppose you are invited to take part in the debate with the motion which says "scientists should find a way of getting rid of all bacteria "Give seven (7) views against the motion.